

SPLUNK ENVIRONMENT SETUP

Objective

Set up and configure a Splunk Enterprise environment and onboard the **BOTSv3 (Boss of the SOC Version 3)** dataset for SPL practice, security log analysis, and future SOC investigations.

Environment

Component	Details
OS	Windows 11
Splunk	Splunk Enterprise
Deployment	Standalone / Local Instance
Dataset	BOTSv3
Index	<code>botsv3</code>
Splunk Web	<code>http://localhost:8000</code>

Installation

Splunk Enterprise was installed and configured on the local Windows system. The Splunk Web interface was successfully accessed through:

`http://localhost:8000`

The Splunk installation directory was identified as `$SPLUNK_HOME`.

The main application directory is:

`SPLUNK_HOME/etc/apps`

BOTSv3 Dataset

The **BOTSv3 dataset** was onboarded into Splunk to provide realistic security data for learning and practicing SPL and SOC analysis.

The dataset contains multiple log and event types. Each log type will be documented and analyzed separately in dedicated sections.

Data Verification

The ingested BOTSv3 data was verified using the Splunk Search & Reporting interface.

Example:

```
index=botsv3
```

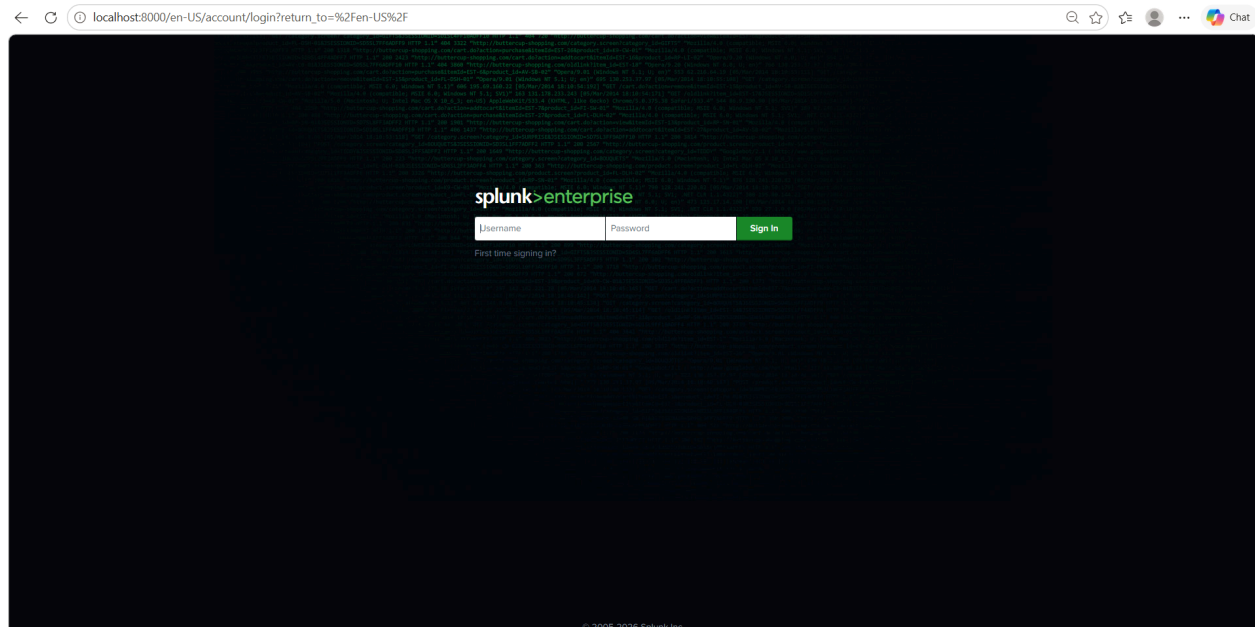
The available data was checked using:

```
index=botsv3  
| stats count by sourcetype
```

This helped identify the different sourcetypes available in the BOTSv3 dataset.

Screenshots

SPLUNK WEB INTERFACE



SEARCH AND REPORTING

splunk> Enterprise

New Search

Search: Time range: All time

103,387 of 91,610 events matched No Event Sampling

Events (103,387) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

1 month per column

Format Show: 20 Per Page View: List

1 2 3 4 5 6 7 8 ... Next

SELECTED FIELDS

- host 23
- source 99
- sourcetype 63

INTERESTING FIELDS

- # date_hour 2
- # date_mday 2
- # date_minute 21
- # date_month 1
- # date_second 60
- # date_wday 2
- # date_year 1
- # date_zone 2
- # index 1
- # linecount 83
- # punct 100+
- # splunk_server 1
- # timeendpos 93
- # timestamp 100+
- # timestamppos 94

937 more fields

localhost:8000/en-US/app/search/search New Fields

Event View:

Time	Event
9/19/19 11:40:50.000 PM	<error><message>Splunkd daemon is not responding: ('Error connecting to /services/shcluster/config/config: [Errno 111] Connection refused',)</message></error>
8/20/18 8:57:09.296 PM	<pre> host = OD-FM-NA-1-Oad2d665d4bdace22.amazonaws.com source = ess_content_importer sourcetype = ess_content_importer [] bytes: 58637 bytes_in: 38 bytes_out: 58687 dest_ip: 172.16.0.178 dest_mac: 82:48:38:1B:83:70 endtime: 2018-08-20T15:27:09.296788Z flow_id: c248fda8-997d-4285-a221-1080976a6bba fragment_count: 0 packets_in: 1 packets_out: 38 protocol: UDP protocol_stack: ip:udp:unknown protoid: 17 src_ip: 13.125.33.138 src_mac: 82:4F:D7:61:53:04 timestamp: 2018-08-20T15:27:09.296118Z tos: 8 version: 4 </pre>

Show as raw text

host = gacruxi-06fe586f3d3c8ce8 | source = streamip | sourcetype = streamip

DATA INGESTION

splunk>enterprise Apps

CyberBlue Messages Settings Activity Help Find

What data do you want to send to the Splunk platform?

Follow guides for onboarding popular data sources

Search bar

Cloud computing
Get your cloud computing data in to the Splunk platform.
10 data sources

Networking
Get your networking data in to the Splunk platform.
2 data sources

OS Operating System
Get your operating system data in to the Splunk platform.
1 data source

Security
Get your security data in to the Splunk platform.
3 data sources

4 data sources in total

Or get data in with the following methods

Upload
files from my computer
Local log files
Local structured files (e.g. CSV)
Tutorial for adding data

Monitor
files and ports on this Splunk platform instance
Files - HTTP - WMI - TCP/UDP - Scripts
Modular inputs for external data sources

Forward
data from a Splunk forwarder
Files - TCP/UDP - Scripts

localhost:8000/en-US

Lessons Learned

- 1] Install Splunk.
- 2] Data Ingestion done using the Botv3 dataset.
- 3] Navigated through all the different components of Splunk (for eg: Search & reporting, Apps etc)

